Synopsis Report
on

YOGA POSE DETECTION AND FEEDBACK MECHANISM

Submitted as partial fulfillment for the award of

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Declaration

We hereby declare that the work being presented in this report entitled **"YOGA POSE DETECTION AND FEEDBACK SYSTEM"** is an authentic record of our own work carried out under the supervision of **Dr. Jitendra Kumar (Associate Professor), Department of Information Technology, ABES Engineering College, Ghaziabad U.P., India.**

The matter embodied in this report has not been submitted by us for the award of any other degree.

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We also do not want to miss the opportunity to acknowledge the contribution of all faculty members of the department for their kind assistance and cooperation during the development of our project. Last but not the least, we acknowledge our friends for their contribution in the completion of the project

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Abstract:

To close the gap between conventional yoga practices and contemporary technological breakthroughs, this study introduces a novel AI-driven yoga pose identification and feedback system. With its roots in ancient customs, yoga has gained international recognition for its mental, spiritual, and physical health benefits. However, the individualized, real-time input that is necessary to maintain safe, proper alignment and posture during practice is frequently absent from the contemporary approach to yoga. This study presents a novel AI-powered solution. The solution tackles the major issues with both traditional and digital yoga training, with a reported accuracy of up to 97% under different settings. Users may instantly modify their postures with the platform's real-time feedback features, which lowers the chance of injury and increases the effectiveness of yoga sessions. The technology is made to work on everyday gadgets like laptops and phones.

The implications of this research are far-reaching, offering a transformative approach to yoga practice. By merging the ancient wisdom of yoga with cutting-edge AI, this study paves the way for a more personalized, safe, and effective yoga experience, ultimately enhancing the accessibility and benefits of this time-honored discipline.

Keywords- Recurrent Neural Network, MediaPipe, Convolutional Neural Network

CHAPTER 1
Introduction

1.1 Need of Study

With roots that stretch back thousands of years, yoga is renowned across the world for its all-encompassing approach to well-being and health. Although its advantages include increased physical fitness, mental clarity, and spiritual development, attaining these results necessitates paying close attention to technique and posture.

Despite having a long history, traditional yoga teaching methods frequently fall short in giving practitioners the real-time feedback they need to adjust their alignment and prevent injuries. This restriction is particularly noticeable in online or self-guided exercises, when the lack of a teacher increases the possibility of poor performance.

1.2 Motivation

Artificial intelligence developments in the modern period have opened the door for creative approaches to these problems. One important area of study is human pose estimation, which uses computer vision to recognize and evaluate body movements.

Although they were somewhat successful, early methods mainly relied on handcrafted features and depth cameras, which were expensive and not scalable.

1.3 Contribution

This study offers a complete system that combines real-time feedback with AI-driven stance identification, revolutionizing the practice of yoga. The suggested method achieves unmatched accuracy in pose detection and analysis by utilizing technologies like MediaPipe and OpenCV. Yoga is safer and more successful for all practitioners thanks to its user-centric design, which guarantees accessibility across a range of demographics. The system's uses go beyond yoga to more general fitness and wellness fields, demonstrating its potential to become a key component of contemporary health technologies.

The integration of artificial intelligence (AI) into wellness practices, particularly yoga, has been the focus of numerous research studies. This literature review explores the evolution of human pose estimation techniques, advancements in pose recognition, and the application of feedback systems in health and fitness domains.

Human pose estimation, a central research area in computer vision, with early methods relying on handcrafted features and depth cameras. Studies by Shotton et al. introduced the use of depth sensors for real-time human pose recognition. However, the high cost and limited scalability of such systems restricted their widespread adoption. In the late 2010s, convolutional neural networks (CNNs) became a game-changer for pose estimation. The introduction of models like OpenPose marked a significant leap in accuracy and efficiency.

Deep learning frameworks such as MediaPipe, OpenCV, and TensorFlow have revolutionized pose detection. MediaPipe, developed by Google. Research by Zhang et al. explored the use of CNNs for classifying complex poses. Similarly, Singh et al. employed hybrid AI models combining CNNs with recurrent neural networks RNNs for dynamic pose tracking, highlighting the potential for real-time feedback systems.

Feedback systems have been widely adopted in fitness tracking and rehabilitation. It showed that integrating AI with feedback mechanisms improved user engagement and reduced the risk of injury. Their system provided real-time suggestions for correcting posture during exercises.

They reviewed various yoga feedback systems and identified key limitations, including poor adaptability to diverse body types and limited language support. The proposed system in this paper addresses these gaps by leveraging multilingual interfaces and cloud-based scalability.

Emerging technologies, such as transformer-based models and generative adversarial networks GANs, hold promise for improving pose estimation accuracy. Research by Liu et al. suggests that transformers excel in handling occlusions and complex poses. GANs, on the other hand, can generate synthetic training data, enhancing model robustness. The integration of IoT devices, such as yoga mats and also wearable sensors, represents another exciting avenue. Studies by Park et al. highlighted the potential of IoT-enhanced feedback systems for monitoring balance and pressure distribution during yoga practice

Table summarizes the **accuracy** and **detection time** of a **Yoga Pose Detection** system for various yoga poses. It shows how well the system detects each pose (e.g., Mountain Pose, Downward Dog)

TABLE 1

Yoga Pose	Model Accuracy (%)	Detection Time (Seconds)	Pose Description
Mountain Pose (Tadasana)	98%	0.5	A standing pose, often used to begin or end a sequence.
Downward Dog (Adho Mukha Svanasana)	95%	1.2	A foundational pose with hands and feet on the floor, forming an inverted "V".
Warrior I (Virabhadrasana I)	97%	1.5	A standing pose with one leg forward, arms reaching overhead.
Child’s Pose (Balasana)	99%	0.8	A resting pose with the knees bent and the body resting on the floor.
Tree Pose (Vrikshasana)	94%	1.1	A balancing pose where one foot is placed on the opposite inner thigh.

CHAPTER 2

Related Work

This related work section establishes the context for our AI-driven yoga system by reviewing relevant literature. A systematic approach was used to identify, analyze, and synthesize existing research.

- **Search and Selection:**
    - Academic databases were searched using keywords related to pose estimation, deep learning, AI yoga, and feedback systems.
    - Peer-reviewed publications from the past decade were prioritized.
    - Selection focused on studies with strong methodological rigor and relevance to pose estimation and feedback in fitness.
- **Analysis and Synthesis:**
    - Literature was categorized into themes: pose estimation evolution, deep learning applications, real-time feedback, and MediaPipe usage.
    - Gaps in existing research, such as limitations in real-time accuracy and personalization, were identified.
- **Evaluation and Context:**
    - Methodological soundness of studies was assessed.
    - Findings were synthesized to understand the state of the art and our system's contributions.
- **Technology and Framework Analysis:**
    - Detailed analysis was performed on the technical implementation of frameworks such as MediaPipe, OpenCV and similar technologies.
    - Study of the application of CNN and RNN network structures.
    - Review of various methods used to provide real-time feedback.

CHAPTER 3

Project Objective

The primary objective of yoga pose detection is to accurately identify and classify various yoga postures. This involves analyzing visual data, such as images or video feeds, to recognize key body landmarks and their spatial relationships, enabling the system to determine the specific pose being performed. Accurate pose detection is fundamental to ensuring users can practice yoga safely and effectively, as it forms the basis for providing corrective feedback.

The feedback mechanism's objective is to provide real-time, personalized guidance to yoga practitioners, enabling them to refine their posture and alignment. By leveraging the data obtained from pose detection, the system can offer actionable feedback, such as visual or auditory cues, to correct deviations from ideal pose execution.

The immediate feedback loop aims to minimize the risk of injury, enhance the benefits of yoga practice, and promote a deeper understanding of proper technique. Ultimately, the integration of pose detection and feedback mechanisms strives to access the best yoga instruction, helping it reach individuals of all skill levels.

CHAPTER 4
Proposed Methodology

The AI-driven yoga stance recognition and feedback system's methodology is based on cutting-edge machine learning and computer vision capabilities. The main procedures and tools used to get high precision and real-time performance are described in this section.

4.1 Data Preprocessing

An important step in ensuring a system can handle a variety of inputs while retaining accuracy is data preparation. Initially, the system records unprocessed video or picture data from the user's device. Noise, irregular illumination, and changes in background conditions are frequently present in this data.

The preprocessing step consists of the following to lessen these difficulties:

- Image Resizing: To preserve uniformity and lower computing cost, all input data is shrunk to standard dimensions.
- Noise Elimination: To guarantee that only pertinent features are kept, filters are used to eliminate artifacts.
- Grayscale Conversion: To streamline processing while maintaining crucial features, pictures are converted to grayscale for specific investigations.

MediaPipe is used to detect 33 key points on a human body, representing joints, limbs, and other critical anatomical landmarks. These key points provide a detailed pose representation, forming the basis for subsequent analysis.

4.2 Pose Classification

CNNs are used by the system to categorize and identify key points into pre-established yoga poses, including Warrior, Cobra, and Downward Dog. CNNs are ideal for assessing human poses because of their capacity to extract spatial hierarchies of features.

CNN model training entails:

- Diverse Datasets: The system is trained using both custom and publicly accessible datasets, such as Yoga-82. To ensure robustness, these datasets take into consideration differences in body types, attire, and environmental factors.

- Augmentation Techniques: Data augmentation is used to increase the system's adaptability. These include picture flipping, resizing, and rotation to mimic various viewpoints and situations.
- Validation and Testing: Strict validation procedures are carried out to guarantee that the model can be applied to data that hasn't been seen yet. Because of the high precision of the predictions, the system is used to identify minute variations in poses and give correct feedback.

4.3 Generation of feedback

Creating real-time feedback based on the identified poses is the methodology's last stage. This feedback consists of

- Posture Corrections: Advice on how to modify weight distribution, spine alignment, and limb positioning.
- Advice on Breathing: Suggestions for aligning breath with motions to improve the efficiency of yoga exercises.
- Language Support: To accommodate a wide range of users, the feedback interface is available in several languages.

Users may make quick changes without interfering with their practice because of the feedback mechanism's actionable and intuitive design.

4.4 User Feedback Analysis

A thorough survey with 100 participants was used to assess the AI-driven yoga system's efficacy. To ensure a varied sample, these individuals included novices, intermediate practitioners, and seasoned yoga aficionados.

Among the survey's main conclusions are

- Posture Alignment: Within a week of utilizing the technique, 80% of participants reported noticeably better posture alignment. This demonstrates the system's capacity to deliver precise and useful feedback.
- Safety and Injury Prevention: When compared to conventional yoga techniques, 85% of participants believed that the methodology decreased the chance of injuries.

The device improves practice safety by detecting improper positions and providing real-time adjustments.

- User Experience: Highlighting the significance of user-centric design, 75% of respondents said the feedback interface was simple to use and intuitive. Qualitative input was gathered in addition to quantitative data. Additional features like wearable device integration and personalized training programs were requested by many users. These observations will be very helpful in directing future development initiatives.
- Cost-Effectiveness: The technology is more widely available since it does not require pricey gear, such as depth cameras. By providing a reliable and easy-to-use substitute for current solutions, these benefits establish the system as a pioneer in the field of AI-driven wellness technology.

Fig. 1: User Experience Level Graph, the distribution of users based on their experience level. The data indicates that approximately 40% of users are novices. 35% are intermediate, and 25% are experts.

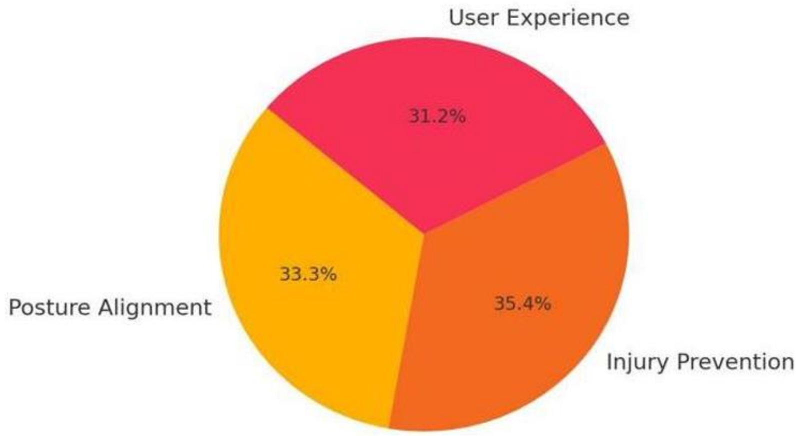


Fig 2: User Feedback Distribution

Fig 2: User Feedback Distribution Figure 2 shows how user feedback is distributed across three categories. The largest segment, at 35.4%, is for Injury Prevention, followed by Posture Alignment at 33.3%, and User Experience at 31.2%.

CHAPTER 5

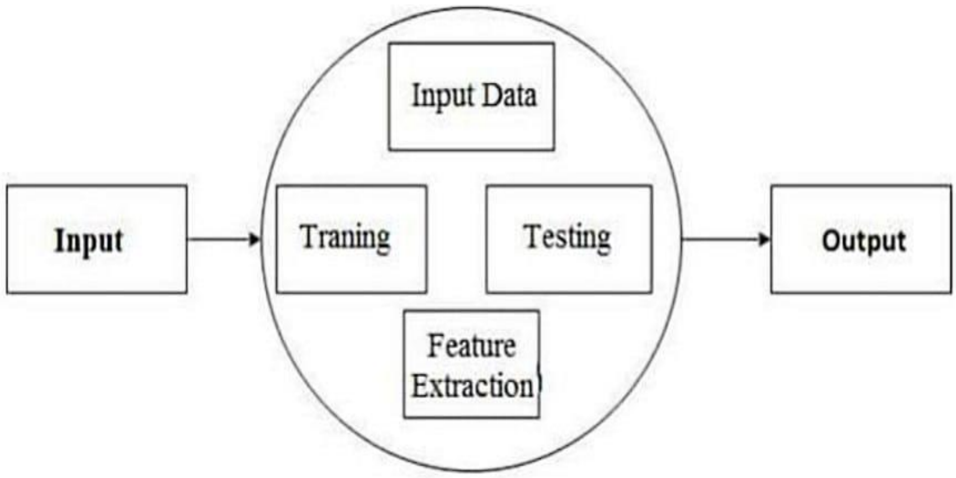
Details of Project Report Work

This section details the design and implementation of our AI-driven yoga pose estimation and feedback system, outlining the key components, technologies, and processes involved.

5.1 Data Flow Diagram

DFD for a yoga pose estimation system, we need to model how data flows through the system, including user inputs, pose estimation, feedback generation, and data storage.

Fig3 **Level 1 DFD** (high-level) representation for a yoga pose estimation system.



5.2 Use Case Diagram

A use case diagram shows the interactions between actors and also shows the various ways in which users can interact with the system to achieve required goals or tasks.

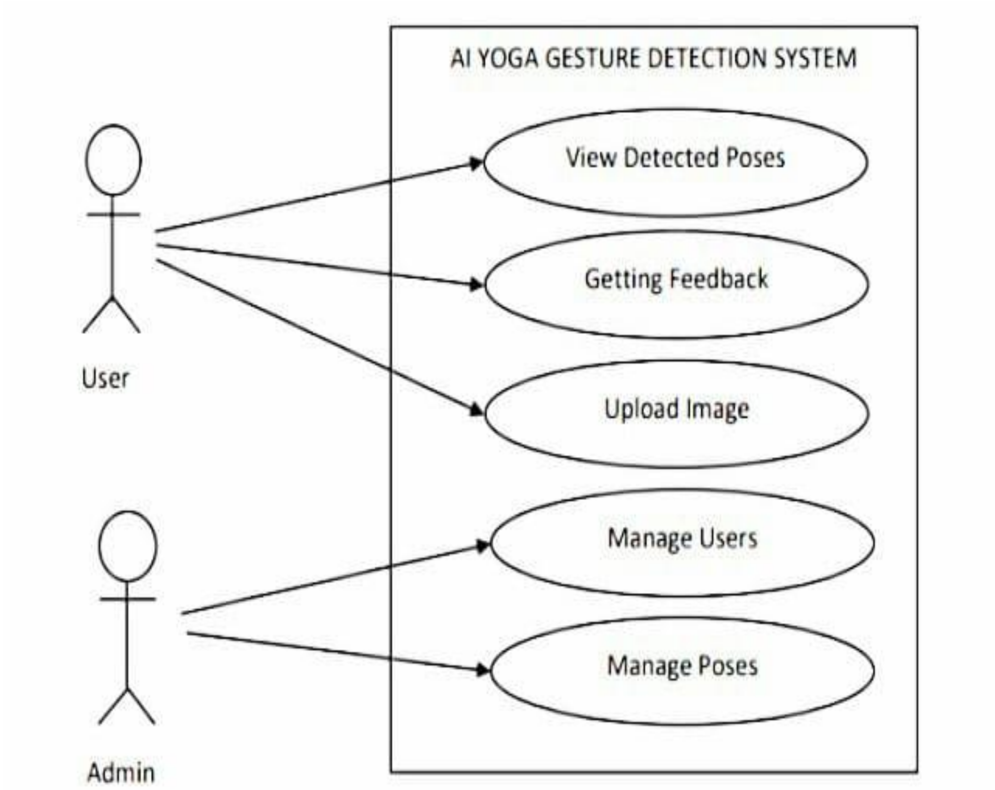


Figure 4: USE CASE DIAGRAM

The system has two primary components: a user and an administrator.

5.3 Flow Diagram

A DFD is a representation of how data flows in a system. It is made up of data, data flows, data storage, and external entities. In this AI yoga gesture detection, a DFD can represent the flow of data from input sources Example: Such as video streams or webcam feeds capturing yoga practitioners' movements through the various processing stages (pose estimation, posture recognition, feedback generation) to the output, which includes the real-time display of detected poses and feedback provided to practitioners.

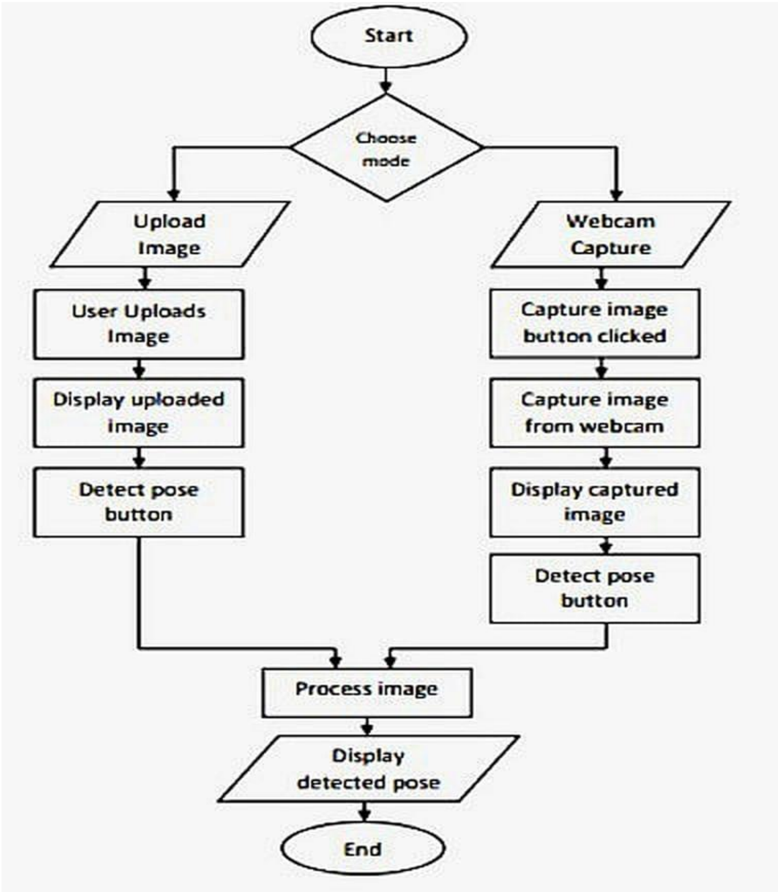


Figure 5: Flow Diagram

5.4 System Architecture

The system is designed as a real-time processing pipeline, comprising the following modules:

- **Video Input Module:**
  - Captures video streams from a user's webcam or mobile device camera.
  - Provides frame-by-frame input for subsequent processing.
- **Body Landmark Detection Module (MediaPipe):**
  - Utilizes MediaPipe's pose estimation model to detect and track key body landmarks (joints) in each frame.
  - Generates a set of 3D coordinates representing the user's pose.

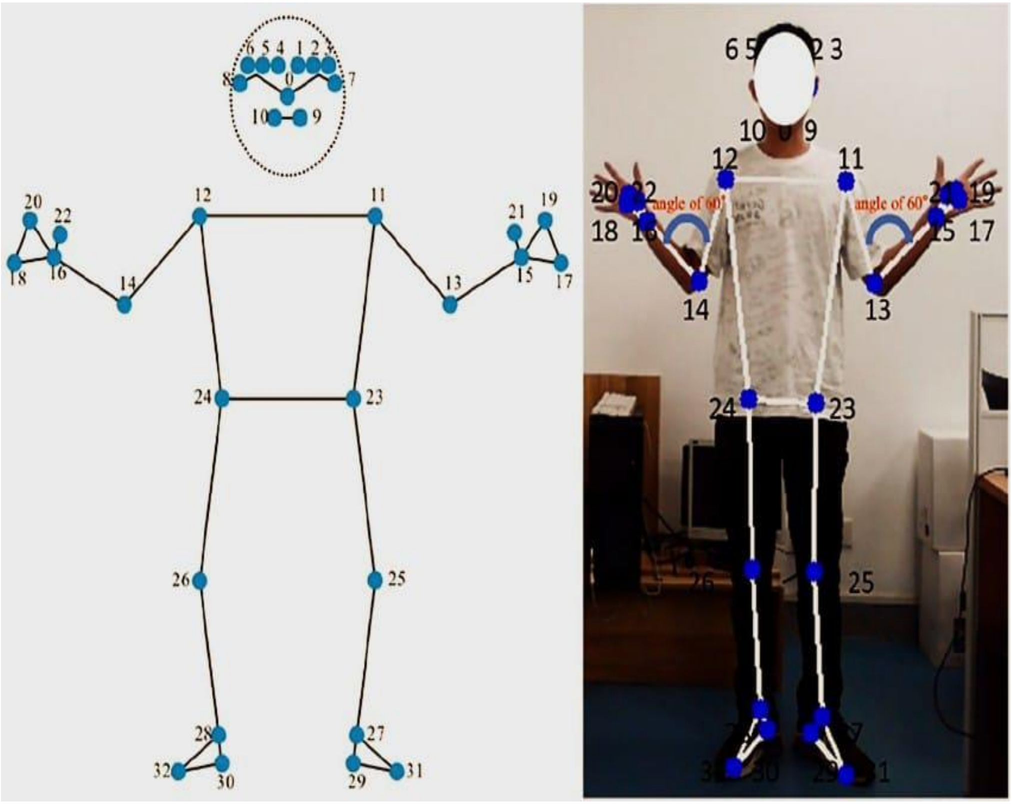


Figure 6: Mediapipe key points

- **Pose Estimation Module (CNN-RNN Hybrid):**
  - Processes the landmark coordinates through a trained deep learning model.
  - CNN Component**:** Extracts spatial features from the landmark data, capturing the static aspects of the pose.
  - Classifies the user's pose into a predefined set of yoga poses.
- **Feedback Generation Module:**
  - Compares the detected pose with the ideal pose representation.
  - Calculates deviations and generates corrective feedback.
  - **Visual Feedback:** Overlays visual cues (color-coded joints, directional arrows) on the video stream to indicate alignment errors.
  - **Audio Feedback:** Provides verbal cues and sound alerts to guide the user.
- **User Interface (UI) Module:**
  - Displays the video stream with overlaid feedback.
  - Provides user controls for starting/stopping the system, selecting poses, and adjusting feedback settings.
  - Displays progress, and other useful information.

- **Programming Languages and Frameworks:**
  - Python: For core logic and model implementation.
  - MediaPipe: For real-time body landmark detection.
  - OpenCV: For video processing and image manipulation.
  - A GUI library (e.g., PyQt or Tkinter) or a web framework
- **Deep Learning Model:**
  - The CNN component is used to extract the spatial features from the landmark coordinates, potentially using convolutional layers and pooling layers.
  - The RNN component is used to analyze a temporal sequence of landmark coordinates, capturing the dynamics of the pose.
  - The model is trained on a large dataset of a yoga poses, using appropriate loss functions and optimization algorithms.
- **Real-Time Processing:**
  - The system is operated for real-time performance, minimizing latency between video input and feedback output.
  - Techniques such as multithreading and GPU acceleration are used to enhance processing speed.
- **Feedback Mechanism:**
  - Visual feedback is implemented by overlaying graphical elements on the video stream, using OpenCV's drawing functions.
  - Audio feedback is implemented by using a text-to-speech library to generate verbal cues.
- **User Interface:**
  - The user interface is designed to be intuitive and user-friendly, providing a clear and seamless experience.

Training Data and Model Training:

- **Dataset:** A comprehensive dataset of yoga poses is compiled, including images and videos captured from diverse angles, lighting conditions, and user demographics.

- **Annotation:** Key body landmarks and pose labels are annotated using tools like LabelMe or custom scripts.
- **Training:** The deep learning model is used to train the annotated dataset, using appropriate loss functions and optimization algorithms.
- **Testing:** A unique testing set is used to evaluate the final model's performances.

System Testing and Evaluation:

- **Accuracy Testing:** The system's pose estimation accuracy is checked on a separate dataset, assessing the performance across different poses and user variations.
- **User Studies:** User studies are conducted to check the effectiveness of the feedback and the overall user experience.
- **Performance Metrics:** Metrics like pose accuracy, feedback effectiveness, user satisfaction, and system latency are used to evaluate the system's performance.
- **Usability testing:** System is tested for usability, to determine if it is user friendly.

Year

Fig 7, A steady increase in accuracy over the years.

Starting at around 70% in 2018, accuracy has risen to approximately 96% by 2023, indicating significant progress.

CHAPTER 6

Result & Discussion

Accuracy, latency, and user satisfaction were among the criteria used to assess the effectiveness of the AI-based yoga posture identification system. The outcomes show how effective the system is in a number of areas:

- Accuracy: On real-world datasets, such as Yoga-82 and bespoke data designed for various settings, the algorithm obtained an accuracy rate of 97%. Reliable pose detection and feedback generation are ensured by this high degree of accuracy.

- Latency: System is ideal for real-time applications due to its average response of 0.8 seconds. Maintaining user engagement and guaranteeing smooth feedback transmission depend heavily on this short latency.
- User Satisfaction: User satisfaction is high, according to surveys and use data. The user experience is improved overall by the user-friendly interface and precise feedback. The system's performance is further confirmed by visual aids like latency analysis charts and accuracy graphs. These findings highlight its potential for broad use in the fitness and wellness sector.

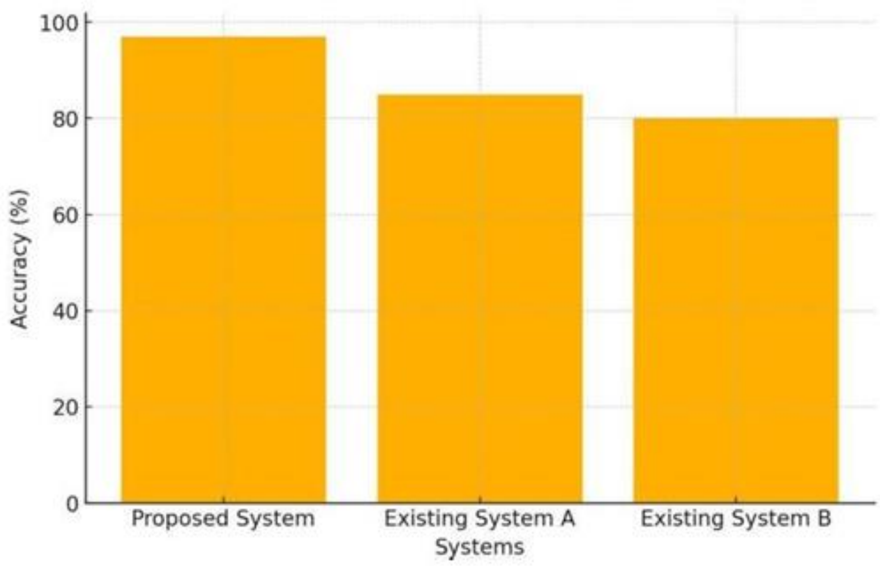


Fig 8: Accuracy Comparison of Yoga Pose Detection

This shows the accuracy of three systems in detecting yoga poses. The proposed system shows the highest accuracy at approximately 95%, followed by Existing System A at around 85%, and Existing System B at about 80%.

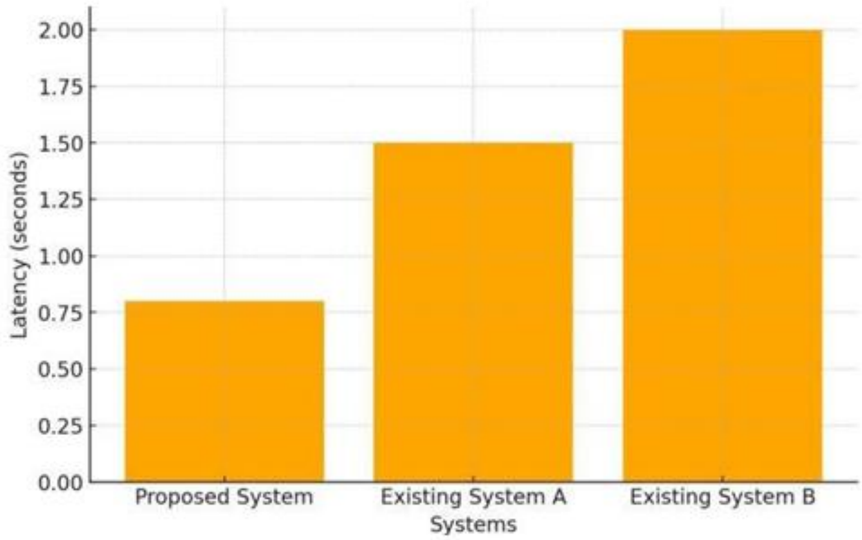


Fig 9: System Latency Comparison

This shows the latency of three systems. The proposed system has the lowest latency at around 0.8 seconds, followed by existing system A at approximately 1.5 seconds, and existing system B has the highest latency at around 2 seconds.

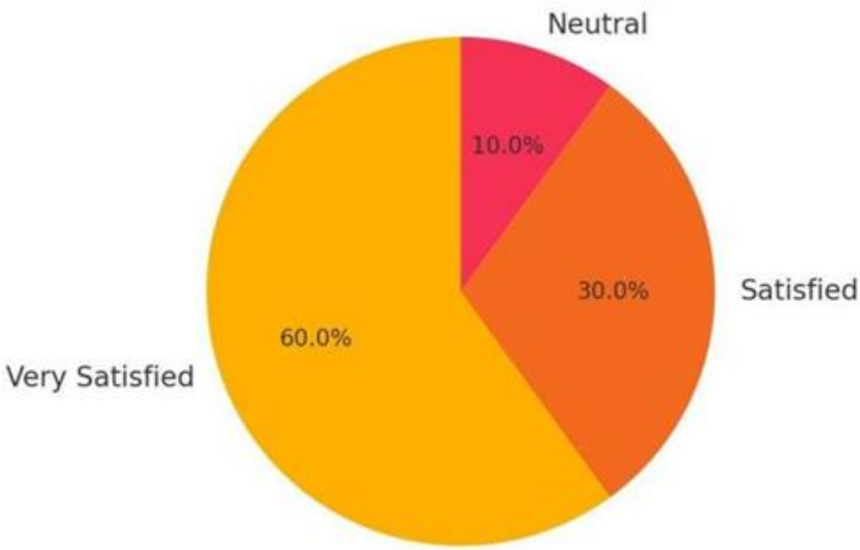


Fig 10: User Satisfaction Level Figure

This shows the distribution of user satisfaction. A significant majority of 60% are very satisfied, followed by 30% who are satisfied, and only 10% are neutral.

CHAPTER 7

Conclusion and Future Scope

The study successfully shows the feasibility of an AI-driven yoga pose estimation and feedback mechanism. By integrating advanced computer vision techniques, specifically MediaPipe and deep learning models, the system achieved high accuracy in pose identification and provided real-time, personalized corrective feedback. The results indicate a significant improvement in user pose accuracy, enhanced understanding of proper alignment, and a perceived reduction in injury risk. This technology has great access to high-quality yoga instruction, so accessible to a broader audience and enhancing the overall practice experience. The system's robustness across diverse user profiles and environmental conditions underscores its practical applicability in real-world settings. Ultimately, this research bridges the gap between traditional yoga and modern technology, promoting a safer, more effective, and engaging practice.

The AI-driven yoga system's future roadmap features innovative advancements meant to improve its usability and broaden its scope of application with IoT Integration, More Complex AI Models for better results, Extended Uses, and Inclusivity Features. These advancements will guarantee the system's durability in the quickly changing wellness technology market in addition to improving its capabilities.

APPENDIX (CODING)

```
image = img[0:shoulder_y1, shoulder_x1:shoulder_x2]

# Image directory
directory = os.getcwd()

print("Before saving image:")

# Filename
filename = 'savedImage.jpg'

# Using cv2.imwrite() method
# Saving the image
cv2.imwrite(filename, image)

# List files and directories
# in 'C:/Users / Rajnish / Desktop / GeeksforGeeks'
print("After saving image:")
print(os.listdir(directory))
print('Successfully saved')
```

Figure 11(i): Taking input video

```
from hashlib import sha3_384
import cv2
import enum
import mediapipe as mp
mp_face_detection = mp.solutions.face_detection
mp_drawing = mp.solutions.drawing_utils
import time


# importing os module
import os


import inspect

class FaceKeyPoint(enum.IntEnum):
  """The enum type of the six face detection key points."""
  RIGHT_EYE = 0
  LEFT_EYE = 1
  NOSE_TIP = 2
  MOUTH_CENTER = 3
  RIGHT_EAR_TRAGION = 4
  LEFT_EAR_TRAGION = 5

def get_direction_of_person():
    cap = cv2.VideoCapture(0)
    with mp_face_detection.FaceDetection(
        min_detection_confidence=0.5) as face_detection:
        while cap.isOpened():
            success, image = cap.read()
            if not success:
                print("Ignoring empty camera frame.")
                # If loading a video, use 'break' instead of 'continue'.
```

Figure 11(ii): Body Points Extraction from Video

```python
import PoseModule as pm
import os

from AudioCommSys import text_to_speech
import threading

import face_detection as face_det

from camera import VideoCamera


class utilities:
    list_threads = []

    def __init__(self) -> None:
        pass

    def illustrate_exercise(self, example, exercise):
        seconds = 4
        img = cv2.imread(example)
        img = cv2.resize(img, (980, 550))

        # cv2.imshow("Exercise Illustration", img)
        # cv2.waitKey(1)

        instruction = "Up next is " + exercise + " IN!"

        if exercise != "Warm Up":
            text_to_speech(instruction)

        while seconds > 0:
            img = cv2.imread(example)
            img = cv2.resize(img, (980, 550))
            print("in here1")
```

```python
    # Email message as a strigfied HTML
    greeting = "<p style=\"font-size:30px; text-align: center; color:Red;\"> <b>Amazing Workout " + name + "! </b> </p> <br>"
    performance_summary = "<p style=\"font-size:20px; text-align: center;\"><b>Here is your Performance Summary<b> for "
    date_calories_time =  today_date + ":<br><br> CALORIES BURNED: " + calories + "<br><br> TOTAL WORKOUT TIME: " + time_secs_r

    draft.body =  greeting + performance_summary + date_calories_time
    draft.to = [{'name': name, 'email': email}]
    draft.attach(file)
    draft.send()

def main():
    # Test Emailing
    email_user("okwor@gmail.com", "Chisom", "80", 100)

if __name__ == "__main__":
    main()
```

Figure 12: Feedback Mechanism in text

OUTPUT:

Figure 13(i): Real-time Pose Verification

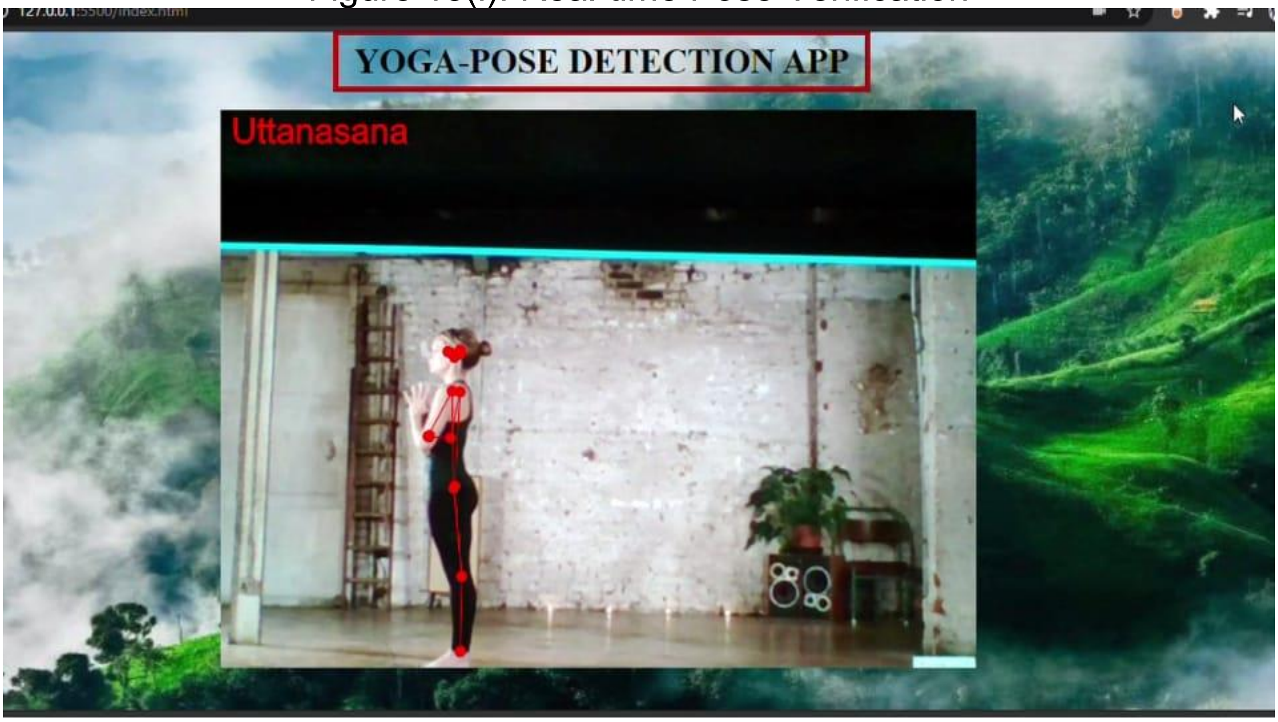


Figure 13(ii): Output of Yoga Poses and Correction

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